

IBM Garage Digital Transformation Case

Electric Industry Digital Transformation
IoT Digital of Productivity Forecast Platform

#IoT #SAAS Platform #Design Thinking Methodology

PROJECT INTRO

PROJECT OVERVIEW

Guangdong Foshan Xinsheng Electric Co., Ltd. hopes to carry out digital transformation through the methodology of IBM garage. To collect production data on the existing equipment by bar code scanning gun. Through the production data to achieve the traceability of production materials and estimate the productive and forecast the capacity in Xinsheng workshop.

RESPONSIBILITIES

Framing Workshop
Design Thinking Workshop
UX/UI Design - Productivity Forecast Platform
Functional Architecture Defined



"This MVP really help NRE to achieve the production process control of the whole process."

IBM GARAGE METHODOLOGY

Co-create

UNDERSTAND

- Explore Business
- Stakeholder Map
- Framing Playback

Co-execute

EXPLORE

- Use Case Sharing
- Empathy Map
- AS-IS Map
- Ideation
- MVP Scope
- DTW Playback

Co-operate

DEFINE

- User Story
- Functional Architecture Design
- Platform Architecture Design
- QA & Training



FRAMING WORKSHOP

EXPLORE BUSINESS OPPORTUNITY

We asked Xinsheng’s leaders and colleagues from each department to list the business opportunities associated with the directions they want to improve. At this activity, the participants list 8 directions after 10 mins discussion. We also asked them to list related character, limitation and challenge of specific direction, business target if achieved. At last we accompany with our clients to vote the most challenging direction they facing.

VOTING RESULT

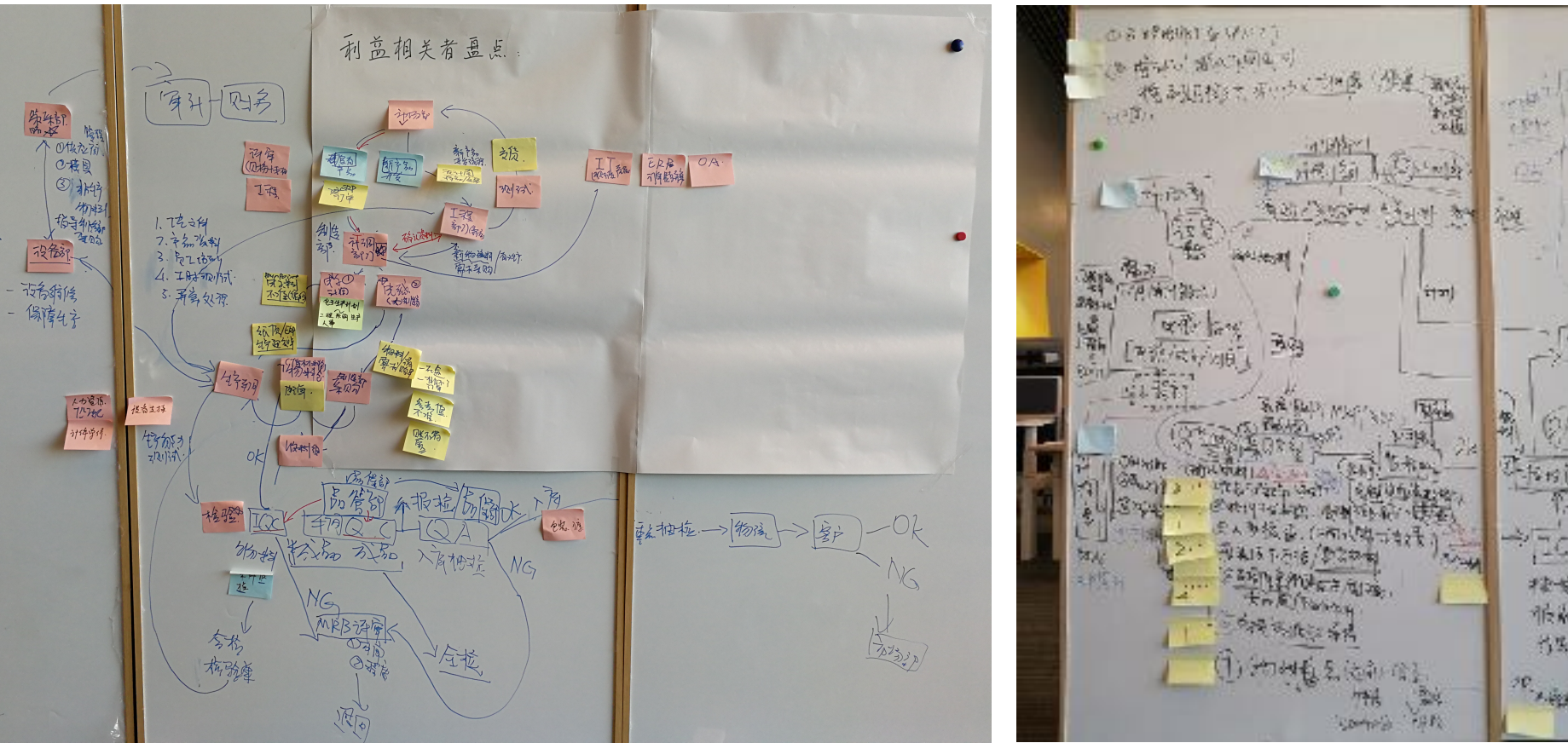
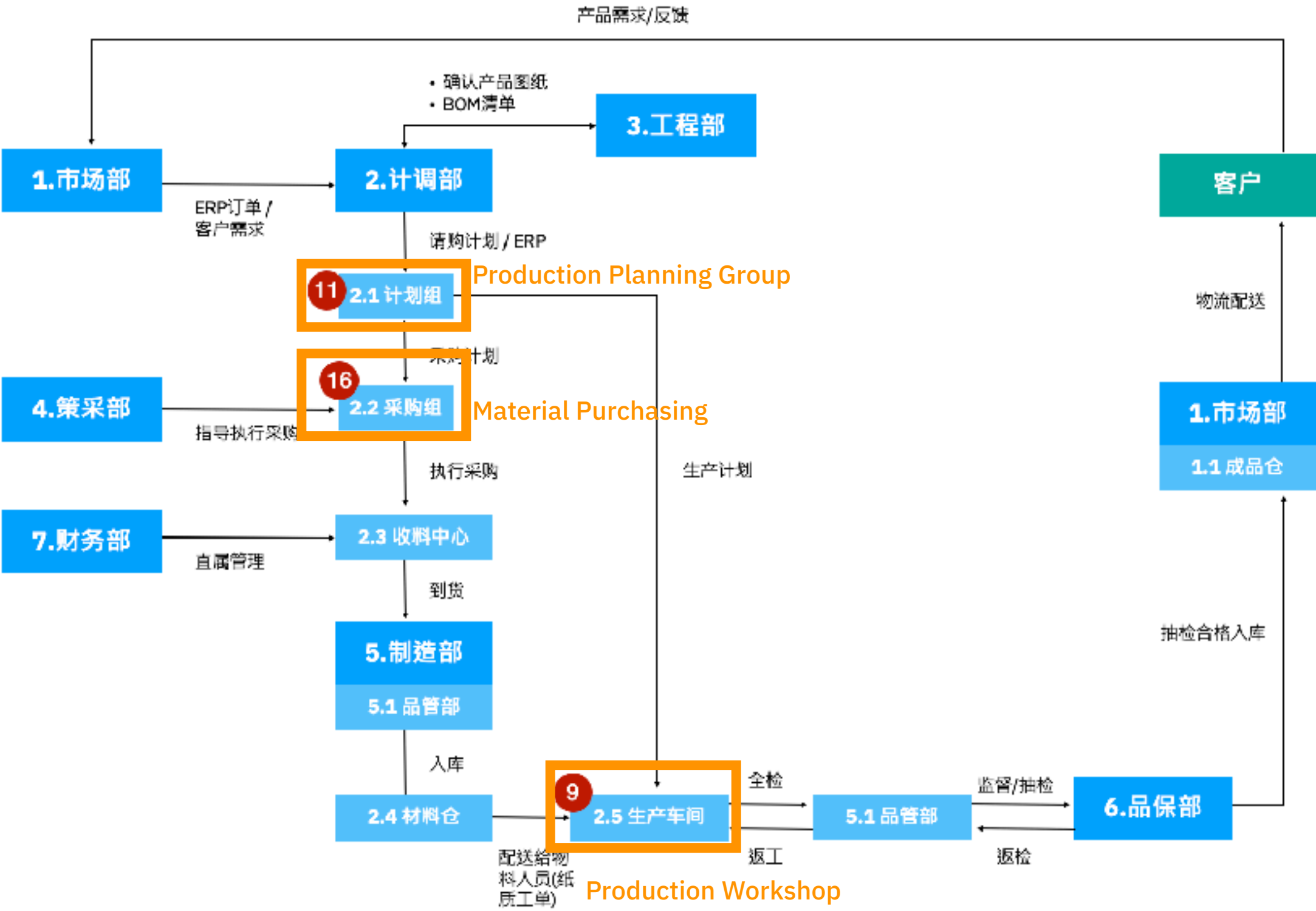
- Product Quality Traceability
- Software Automation / Informatization
- Productivity Workflow Control

业务方向	相关角色	限制与烦恼	业务目标	投票 (重要性)
1. 设备自动化升级	设备部、IT部、生产车间、生产工程部、决策层	设备故障频发、员工不稳定、优化工艺、废气收集	设备自动化程度60%	10
2. 质量追溯	生产人员、QC、工程部、品管部和品保部	客诉2%过高、QC检验数据统计难、投送产品质量问题、新产品导入慢	质量追溯 (100%质量跟踪追溯到客户)	5
3. 软件自动化/信息化	RD和IT部、生产所有部门	制图和BOM表导入过长、公司需要更多支持力度、不停跑问、不能按时生产和产能问题	软件自动化,管理信息化 [计划、排产、交货] (结合物料周期、产能、系统自动分配交期,生产按此执行)	15
4. 流程控制	生产制造、策采部、计划部、质量控制部、销售部、人事部	产能问题、产品定制化多、订单多效率低、库存压力、物料不足、人力资源问题	生产流程：按时交货 按客户要求交货 全流程控制	5
6. 标准	市场部	定制品多、客户变更率高	明确产品开发方向	6
8. 管理合作	决策层、各部门	员工积极性不高不主动	员工稳定	7

Results of Clarifying Business Direction

1. Equipment automative upgraded (Physical equipment problem can not be solved by software development)
2. Quality tracking
3. Software automation / informatization
4. Productivity workflow control

STAKEHOLDER MAP



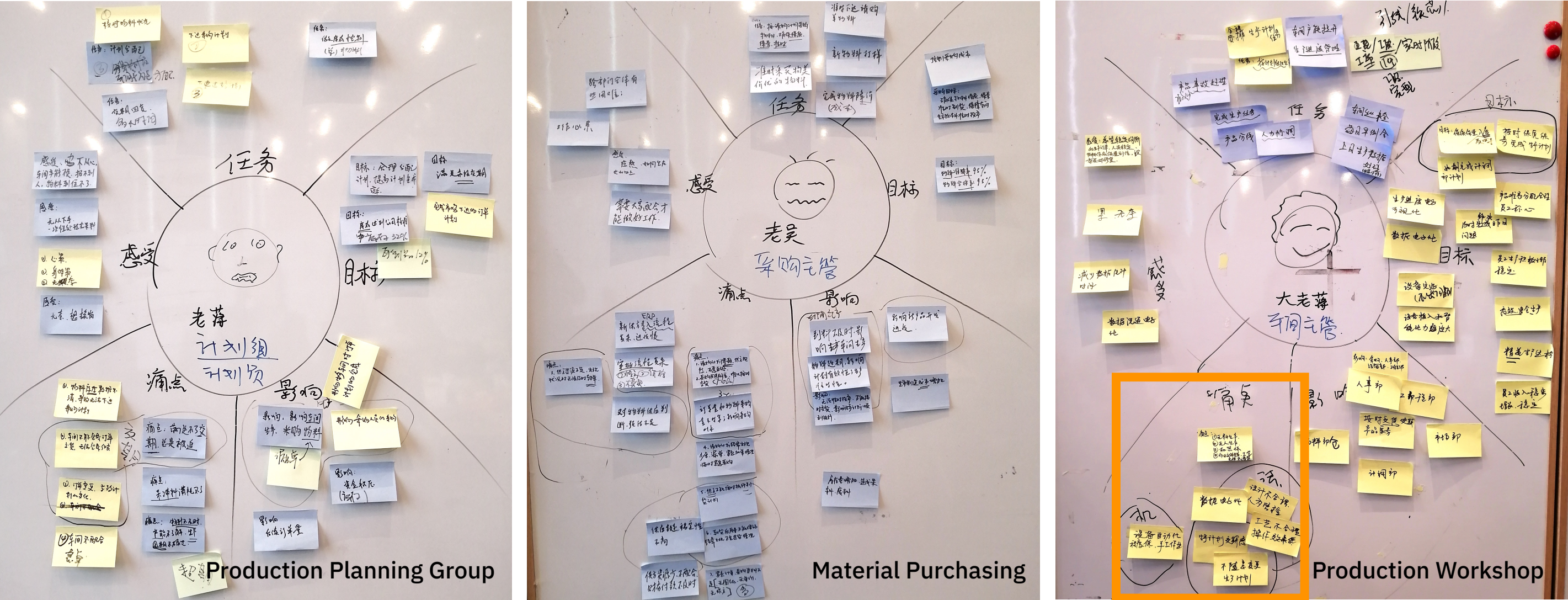
Find The Key Stakeholders To Join Design Thinking Workshop

1. Production Planning Team
2. Material Purchasing Team
3. Production Workshop (include 3 teams)

Based on business direction, we need to invite key person to join next workshop to understand their workflow, pain points and what they think in daily job.

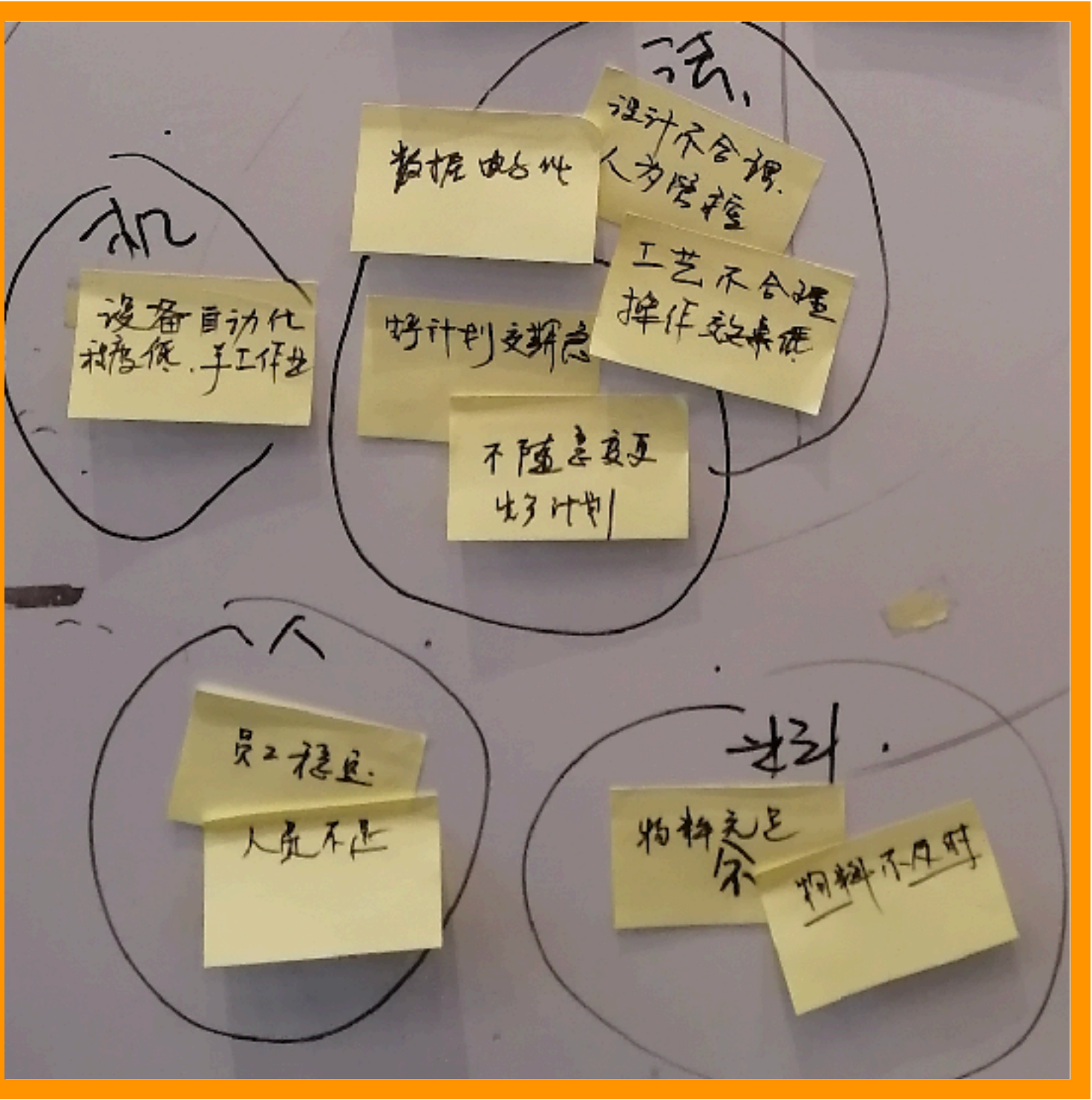
DESIGN THINKING WORKSHOP

EMPATHY MAP



Pain Points From Production Workshop

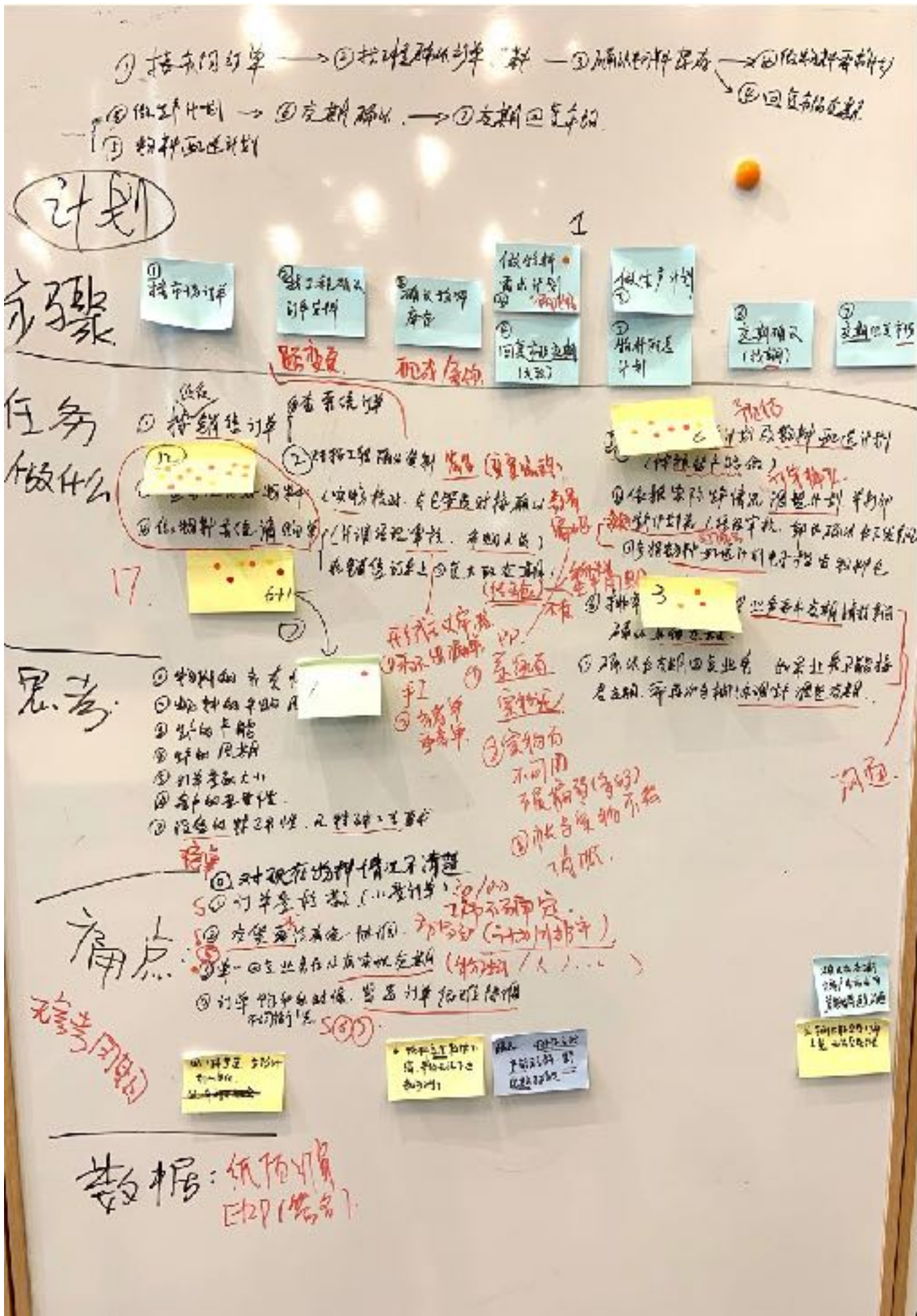
1. Equipment : low-level automation
2. Employees : without stable and sufficient labor
3. Data : without production data to refer
4. Raw Materials : without stable supplement



Empathy Map

We had invited 3 groups of stakeholder to join second workshop (Design Thinking Workshop). First Step I designed is empathy map to have understanding of our users.

AS-IS - PRODUCTION PLANNING GROUP



Key Task

步骤	接市场订单	找工程确认 订单资料	做物料需求计划 / 回 复市场交期 (大致)	找工程确认订单 资料	做生产计划 / 物流配送计划	与生产车间 交期确认	交期回复市场
任务	接收销售订单 物料系统请购单	对接工程确认 资料	查系统库存资料(17)	做物料系统购 单	规划生产计划(预估 产能)及物料配送计 划(订单排程)(6)	排单后车间根据业 务要求交期,并确 认交期时间(3)	回復业务交期,如 不能接受,需再次 与车间协调并调整 交期
思考	• 物料齐套性 • 物料的采购周期 • 生产的产能 • 生产的周期		• 订单量的大小 • 客户的重要性 • 产品的特殊性、特殊工艺的要求				
痛点	• 检验对现存物料情况不清楚 • 订单量零散(小量订单) • 交货需求没有统一协调 • 单一回复业务没有实现交期 • 订单饱和时,众多订单很难协调 • 订单变更时,影响原有生产计划 • 物料库存不清楚,无法下达采购计划 • 生产不及时、产能不了解、生产周期不确定						

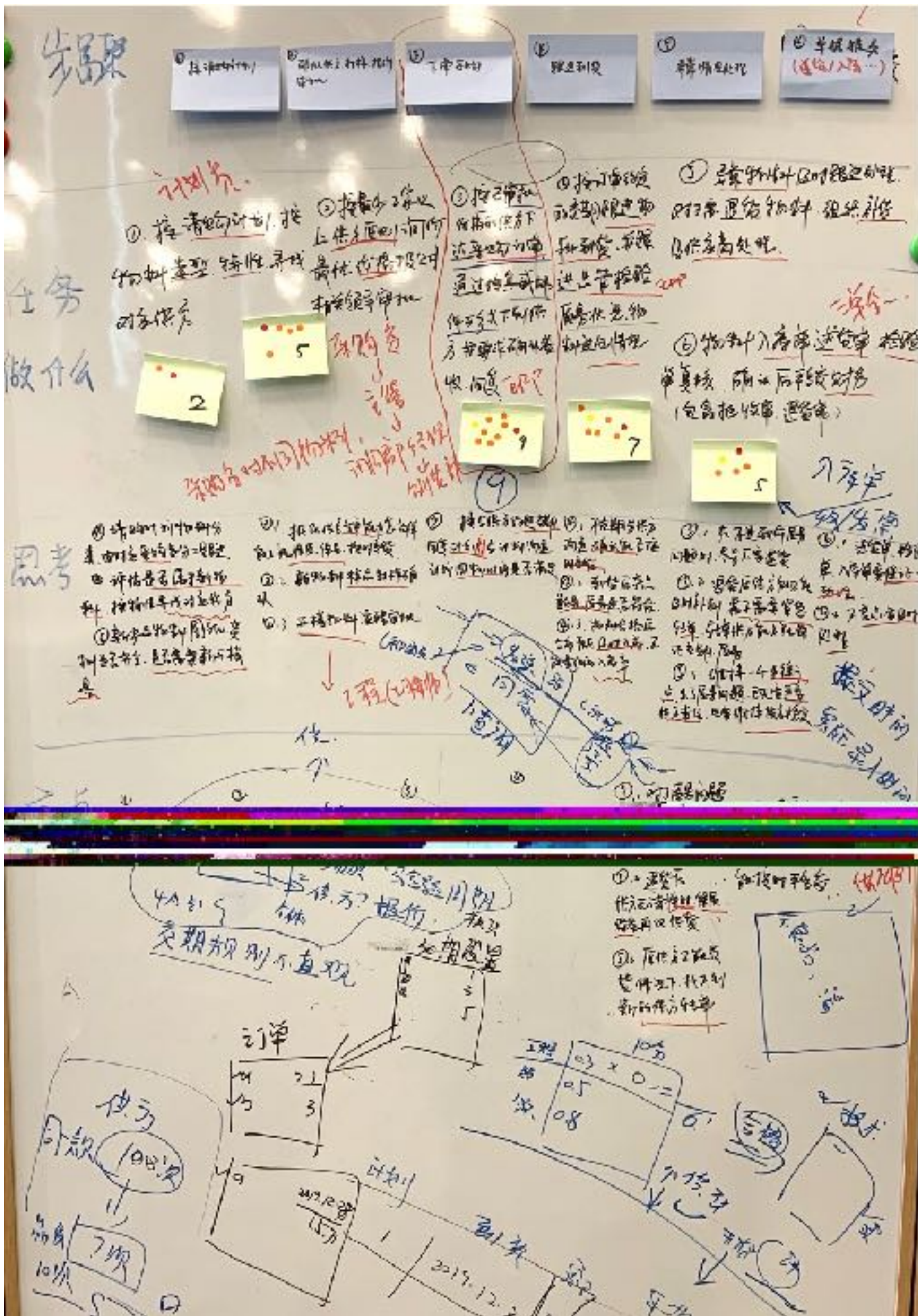
Key Tasks In Planning

1. Purchasing Plan : care about delivery time and feedback.
2. Production & Logistics Plan : evaluate productivity and order arrangement.
3. Production Workshop : confirmation of delivery Time.

Pain Points

1. Do not know the current inventory that affect purchase plan.
2. Difficult to compromise orders changing and deliver time.
3. Difficult to coordinate labors and production materials.

AS-IS - MATERIAL PURCHASING



Key Task

步骤	接请购计划	确认供方、打样、 报价审批 5	下单采购 9	跟进到货 7	异常情况处理	单据提交 (进货 / 入物) 5
任务	接请购计划, 按照物料类型、特性寻找对应供方(2)	最少 2 家进行询价, 以最优价格(5)	按已审批价格的供方下达采购订单, 通过传真或由件形式吓到工坊, 并要求确认签收回复(ERP)(9)	按订单约定的交期跟进物料到货, 并跟进拼箱检验质量状况, 物料进仓状况(7)	异常物料即时跟进处理, 对于需要退货物料、组织补货及供应商处理	物料入库单、送货单、检验单复核, 确认后移交财务 (包含拒收单、退货单)(5)
思考	- 请购计划物料分类, 由对应采购员分工跟进 - 评估是否属于新物料, 按特性寻找对应供方 - 新产品物料图纸资料是否齐全, 是否需要新开模具	- 评估报价供方生产能力: 保质、保量、按时交货 - 新物料样品对样确认 - 与工程部的工程师进行开模物料流程审批	按与供方交期回复, 与计划部沟通计划回料时间是否满足	- 不定期与供方沟通确认能否按时交货 - 到货后产品数量、质量是否符合 - 物料合格后, 仓库有无即时入库, 不能影响入库率	- 在不是致命质量问题时, 劲量不要退货 - 退货后供方能否即时补料, 需不需要紧急转单, 转单供方保证交期与质量 - 维持一个平衡点, 出了质量问题既要追究供方责任, 也要维持供方稳定	- 送货单、检验单、入库单要保证一致性 - 不良品要及时处理
痛点	请购计划零散、供方抱怨不愿配合	采购周期短, 难以按时交货	- 供方资源不足, 出现状况时无法及时转单 - 紧急订单、采购资料不足, 无图纸、无单价、无供方	到货后质量不能保证, 经常出现不良退货情况	- 送货单、检验单、入库单要保证一致性 - 不良品要及时处理 - 对于质量问题不是致命性的评审意见不符 - 退货后供方无法按时保质保量再次供货 - 原供方不能交货情况下, 找不到新的供方转单	不良品不能及时请退, 单据积压不能按时移交供应商

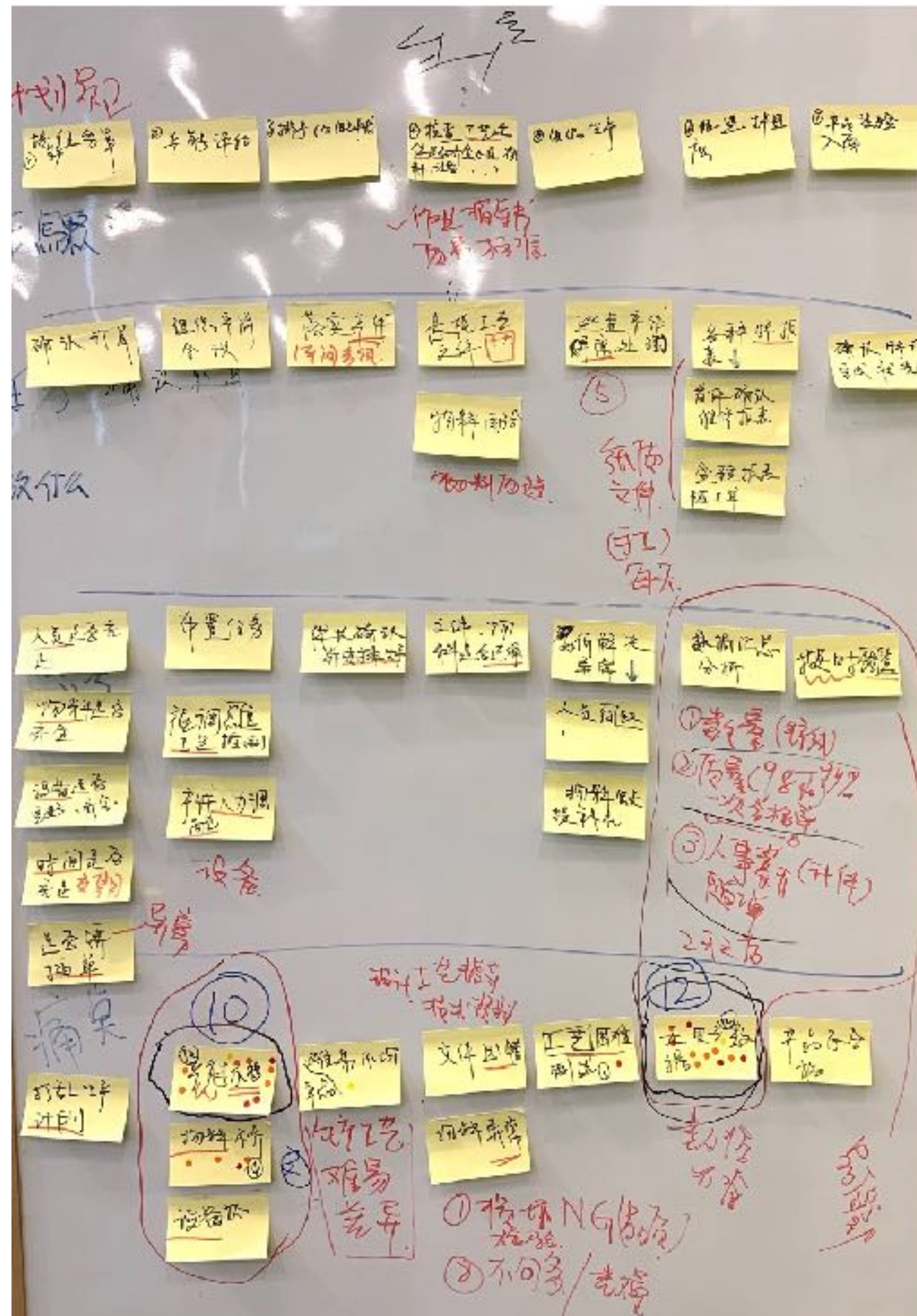
Key Tasks In Purchasing

1. Purchasing Price : review & approve actual quotation.
2. Place The Order : after compared quotation.
3. Delivery Of The Goods : follow up & exam.
4. Transfer To Warehouse : store the inventory

Pain Points

1. Supplier complain purchasing plan are scattered.
2. Material shortage & deliver delay.
3. Quality inspection is not careful enough.

AS-IS - PRODUCTION WORKSHOP



Key Task

步骤	接生产任务单	产能评估 14	排产分配平衡	检查工艺文件是否齐全 (物料、设备..)	组织生产	跟进生产进度 12	产品检验入库
任务	确认订单	组织产前会议	落实产能	悬挂工艺文件、物料配给、物料配送	巡查文件 (每天、纸质、异常处理)	- 各种生产报表 - 首件确认、维修报表 - 全检报表、随工单	确认生产订单完成状况
思考	- 人员是否充足 - 物料是否齐全 - 设备是否完整齐全 - 时间是否充足交期 - 是否需要插单	- 布置任务 - 强调重点工艺控制 - 产线人力调配	产线确认并安排生产	文件、物料是否正确	- 如何解决异常 - 人员短缺 - 物料缺损补充	- 数据汇总分析 - 数量统计 - 质量(一次合格率达到 98%) - 人事薪酬随工单 (两天之后)	按时检验
痛点	打乱生产计划	- 产能不足(10) ● - 物料不齐(4) ● - 设备老旧	产品生产工艺难易差异(1)	- 设计工艺指导、技术资料出错 - 物料异常	工艺製造困难	无电子数据(12) ●	产品不合格

Key Tasks In Production

1. Evaluate Productivity : organize the meeting
2. Follow Production Schedule : review report and production of orders.

Pain Points

1. Production capacity is insufficient.
2. Materials are not ready.
3. The production equipment are too old.
4. Lack of digital data to refer.

IDEATION - PRODUCTION PLANNING GROUP

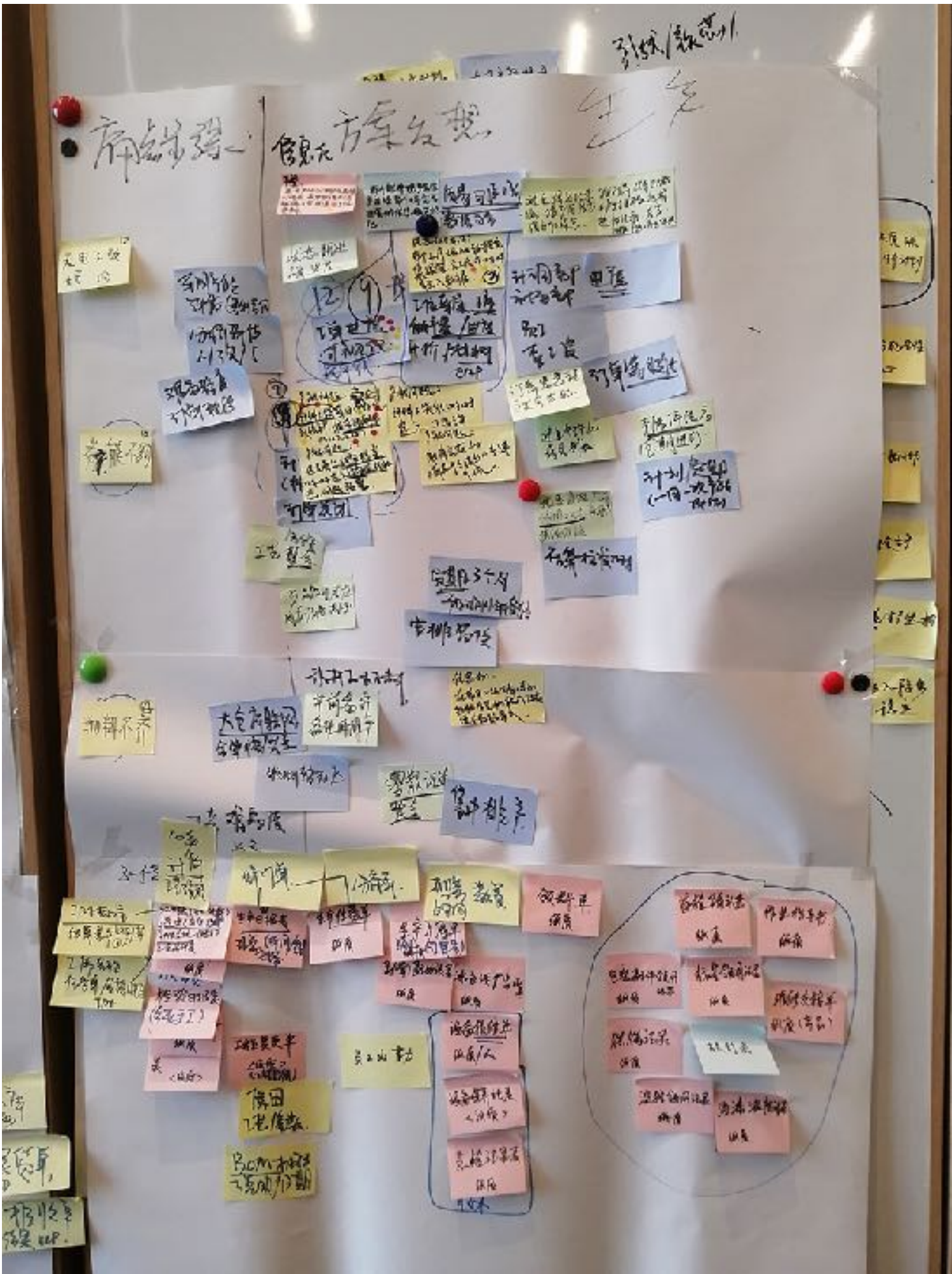


主要痛点	方案描述	票数
1. 对现存物料情况不清楚 1 Do not know the current inventory that affect purchase plan.	制定完整的计划、分析系统、生产计划分析表、材料周期、产能订单特性、客户需求、产品特性 物料(主仓)查询系统，有现存、交货周期、物料出帐入帐可视化、可快速查询 产线物料清晰(线边仓数字化) ● 建立一个物料材料分拆处理模块，不是现在有物料需求的时候 ● 各物料基础资料完善，如订购量、交期 安全库存提醒(日常)预警	(12) ★
2. 交货需求没有统一协调(产能不准) 2 Difficult to compromise orders changing and deliver time.	产能数据准确、清晰 ●	(10) ★
3. 订单变更时，没有实现交期	订单发生变更时，系统可提供预警实时掌握	(3)
4. 订单不均衡，时忙时闲	订单进度可视化，完工后可输入产品数量 ●	(1)
5. 订单量零散(小量订单)	横向排单优先 ●	(0)
6. 产品特性不同	原料不够集中 ●	(0)

Design Solutions for Pain Points

11. Digitalization of inventory stored in temporary warehouse.
21. Production statistics are more accurate with orders, process and numbers of product.
22. Visualized materials issues for each production line.

IDEATION - MATERIAL PURCHASING



主要痛点	方案描述	票数
1. 到货后质量不能保证，经常出现不良退货情况(不良率频次)	• 供方评级考核制度，体现供方物料品质、出现问题频次并加以分类	(3)
2. 交期规则不够直观，影响交期设定	• 供方考核及支付数据收集模块(需入库单、发票关联、提交时间、实际录入时间)	(9) ★
3. 缺少供方考核管理	• 供方管理模块：	(3)
4. 供方资源不足，出现状况时无法即时转单	1. 全检供方	
5. 请购计划零散、供方抱怨不愿配合	2. 特殊供方	
6. 采购周期短，难以按时交货	3. 其他供方供给	(2)
7. 缺少内部采购人员岗位考核	• 自动竞价系统	
8. 缺采购订单监督可视平台	• 查询采购订单：发送供应商/回复，在平台中可优先查询且唯一	(6) ★
9. 紧急订单、采购资料不足，无图纸、无单价、无供方	暂时无方案	

- 1 Quality inspection is not careful enough.
- 2 Material shortage & deliver delay.

Design Solutions for Pain Points

11. Collect the payment data : record inventory, pay check, offer time and record time.
21. To check order on the platform and sent the message to supplier.

IDEATION - PRODUCTION WORKSHOP



主要痛点

1. 无生产过程（电子）数据：
无法及时计件，各工序过程无法追溯

1 Lack of digital data to refer.

2. 产能不够：

(1) 无工艺难易评级，造成按件计酬机制不公平、产能评估不准

2 Production capacity is insufficient.

(2) 人员流动性高

(3) 排产计划不稳定

(3) 设备陈旧造成产能低下，未及时提供预防性维护

3. 物料不齐：

生产前未备齐物料，无法提前得知物料信息

方案描述

• 实时计件，每月自动统计薪酬
• 每道工序可追溯

票数

(12) ★

• 按订单优先级和权重系数，实现智能排产，并可追踪订单更改痕迹
• 对设备实现故障修和预防性检修

(9) ★

• 产前备齐物料再开工，并整合零散订单

(0)

Design Solutions for Pain Points

- 1
1. Real-time salary calculation & summarize per month.
 2. Planning sheet can be tracking.

- 2
1. Intelligent priority of orders and tracking the change.
 2. Predicted equipment broken and real-time alert.

DESIGN THINKING WORKSHOP

IDEATION SUMMARY

After ideation activity, we discussed with clients to choose the MVP scope. So we summarize the result of this workshop as below list to help clients to vote.

MVP Scope

Production Planning

★ MVP : Intelligent supply Chain

1. Digitalization of inventory stored in temporary warehouse.
2. Visualized materials issues.
3. Demo statistics (orders, process and numbers of product).

Material Purchasing

★ Second MVP

1. Record inventory, pay check, offer time and record time.
2. To check order and feedback to supplier in real-time.

Production Workshop

★ MVP : Intelligent production

1. Real-time salary calculation & summarize per month.
2. Track planning sheet.
3. Intelligent priority of orders and tracking the change.
4. Predicted equipment broken and real-time alert.



Invited manufacturing experts to exchange ideas



DEVELOPMENT ROADMAP

Phase 1

Phase 2

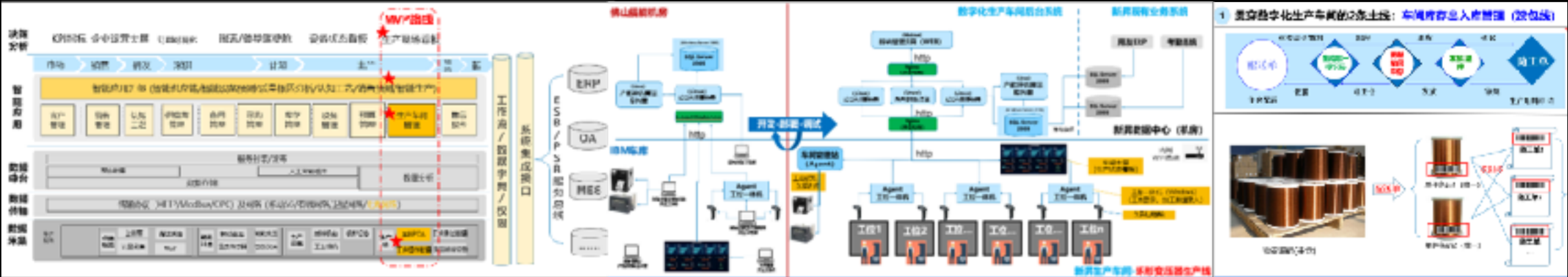
Understanding Existing Systems & Data



Systems Understanding

Data Analysis by DS

Visualize Data & Workflow

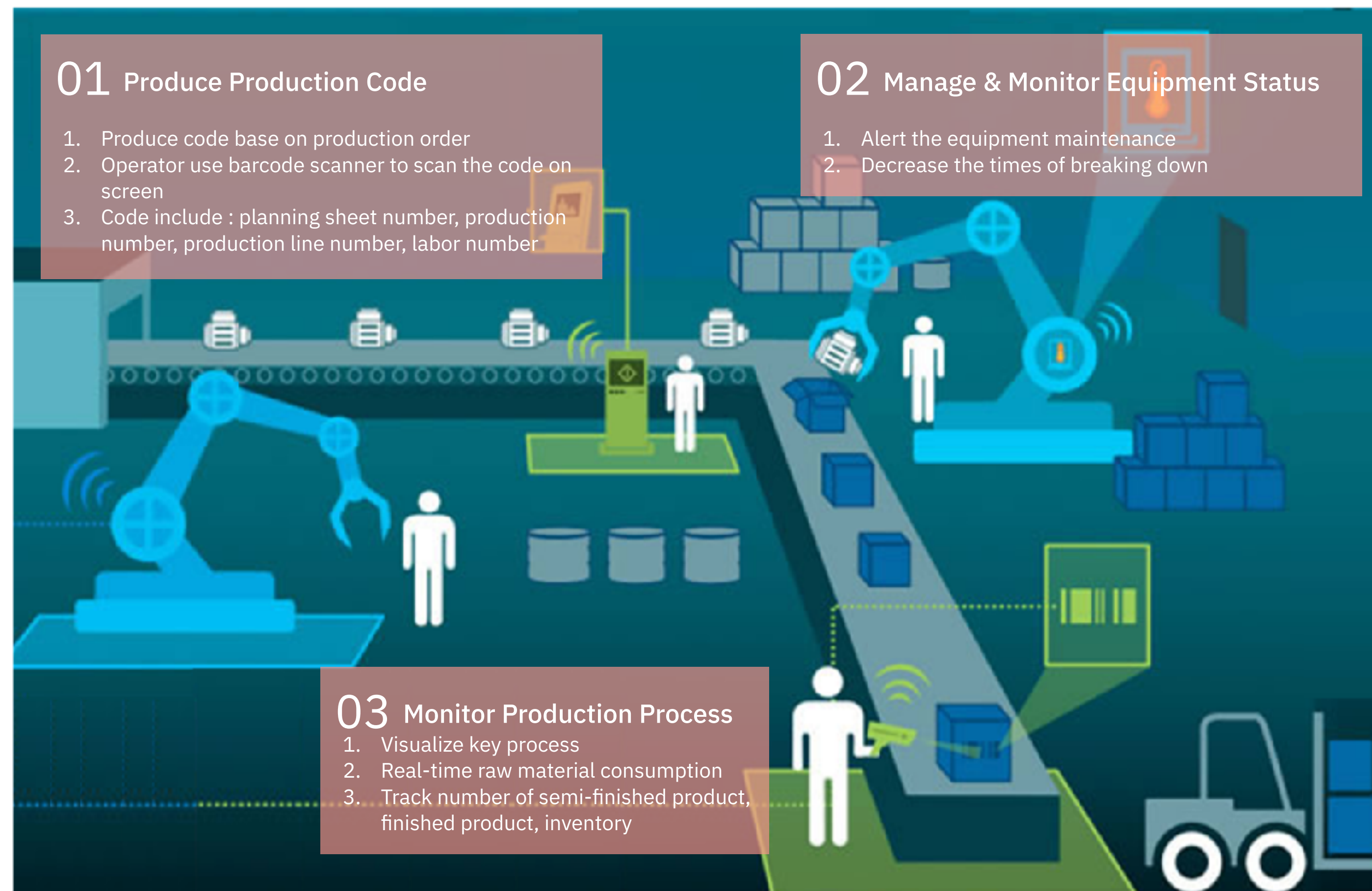


Architecture Design

Platform Architecture Deployment

Raw Material Solutions

SIMULATE PRODUCTION WORKSHOP SCENARIO



Simulate production workshop scenario

Production Orders

1. Confirm production orders

Arrange Orders & Productivity Evaluation

1. Assign tasks
2. Control manufacturing process
3. Arrange labors on specific production line
4. Confirm & prepare production line

Check BOM Sheet

1. Confirm docs and raw materials
2. Hang docs
3. Supplement delivery

Organize Production

1. Checklist confirm
2. Raw material supplement

Exam Inventory

1. Confirm completed status of orders

PLATFORM LANDING PAGE



IBM

IBM Cloud Garage © Copyright IBM Corporation 2020.

- 1

Workshop team leader can monitor the real-time productivity of each line.
- 2

Real-time display ongoing orders and monitor degree of completion.
- 3

We provide a small tool for calculating labors and estimate time of completion
- 4

If you use the tool, the table could demonstrate how many orders will be influenced.

MEMBER MANAGEMENT PAGE

NRC

资源管理

人员

用户

角色

权限

PLM 管理

生产管理

库存管理

资源管理 / 用户

新增

删除

1

☐	序号	员工号	姓名	部门	身分证	电话	创建时间	操作
☒	1	00001	林小华	生产製造部	112233445566	15750001234	2020-01-20 12:00	查看条码 重设密码 查看角色
☐	2	00002	王小明	生产製造部	112233445566	15750001234	2020-01-20 12:01	查看条码 重设密码 查看角色
☐	3	00003	陈大大	采购部	112233445566	15750001234	2020-01-20 12:02	查看条码 重设密码 查看角色
☐	4	00004	张三	生产製造部	112233445566	15750001234	2020-01-20 12:03	查看条码 重设密码 查看角色
☐	5	00005	林大明	生产製造部	112233445566	15750001234	2020-01-20 12:04	查看条码 重设密码 查看角色
☐	6	00006	吴小英	市场部	112233445566	15750001234	2020-01-20 12:05	查看条码 重设密码 查看角色
☐	7	00007	陈雪雪	生产製造部	112233445566	15750001234	2020-01-20 12:06	查看条码 重设密码 查看角色
☐	8	00008	林美美	采购部	112233445566	15750001234	2020-01-20 12:07	查看条码 重设密码 查看角色
☐	9	00009	陈三	市场部	112233445566	15750001234	2020-01-20 12:08	查看条码 重设密码 查看角色
☐	10	00010	李四	生产製造部	112233445566	15750001234	2020-01-20 12:09	查看条码 重设密码 查看角色

总共 : 201

1

2

3

4

...

20

前往

2

页

每页显示

10

条

Member list

1

2

3

Admin can manage employees and assign the unique code for each member. We also synced up with CRM system to save prevent discord of member list. Employee can use barcode scanner to login-in in production workshop.

选择人员

☐

序号

员工号

姓名

部门

身分证

电话

创建时间

搜寻...

搜寻...

搜寻...

搜寻...

搜寻...

搜寻...

🕒

☒	1	00001	林小华	生产製造部	112233445566	15750001234	2020-01-20 12:00
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☒	2	00001	林小华	生产製造部	112233445566	15750001234	2020-01-20 12:00
☒	1	00001	林小华	生产製造部	112233445566	15750001234	2020-01-20 12:00
☒	2	00001	林小华	生产製造部	112233445566	15750001234	2020-01-20 12:00
☐	2	00001	林小华	生产製造部	112233445566	15750001234	2020-01-20 12:00

总共 : 201

1

2

3

4

...

20

前往

2

页

每页显示

10

条

查看条码

林小华

3



A12345678A

关闭

取消

确定

Batch Edit

SCENARIO IN PRODUCTION WORKSHOP

Workers use bar code scanner to login in by scanning the code on badge

Scan the code on production task list paper, then confirm the info on the screen and start to work.

Also use scanner to scan the raw material bucket, system will automatically record how many raw material consumed

Pause or completed an task, just need to click the button on touch screen.

If need to logout, just scan the badge again.

Before production

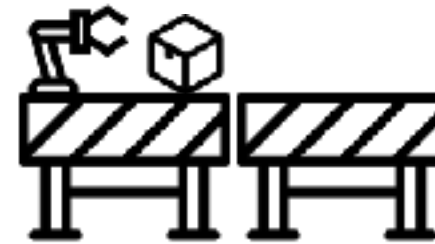
Before production



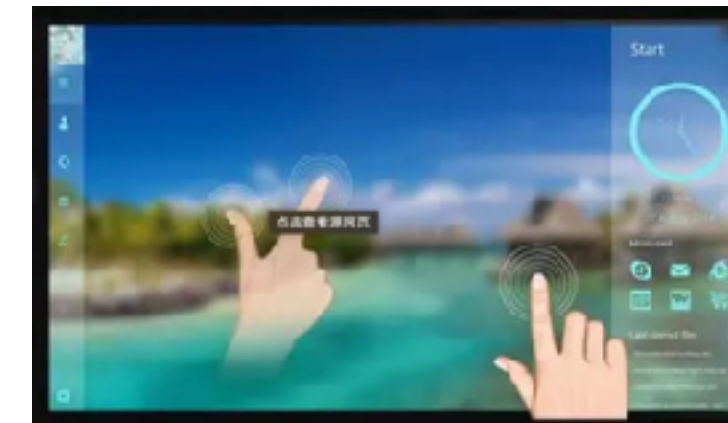
Workers use bar code scanner to login in by scanning the code on badge



Scan the code on production task list paper, then confirm the info on the screen and start to work.



Also use scanner to scan the raw material bucket, system will automatically record how many raw material consumed



Pause or competed an task,
just need to click the button
on touch screen.



If need to logout,
just scan the badge
again.

Before production



Before production

[illegible][illegible]



INVENTORY CONSUMPTION



Each iron cycle has unique bar code and bound with inventory.



After scanned iron cycle, this material will automatically bound with production task list.



After completed production task, inventory will be automatically deducted.



Dashboard will display real time productivity and inventory statistics.



Before production

Before production

随工单工具

随工单号: S20230202020

产品编号: P-212-224

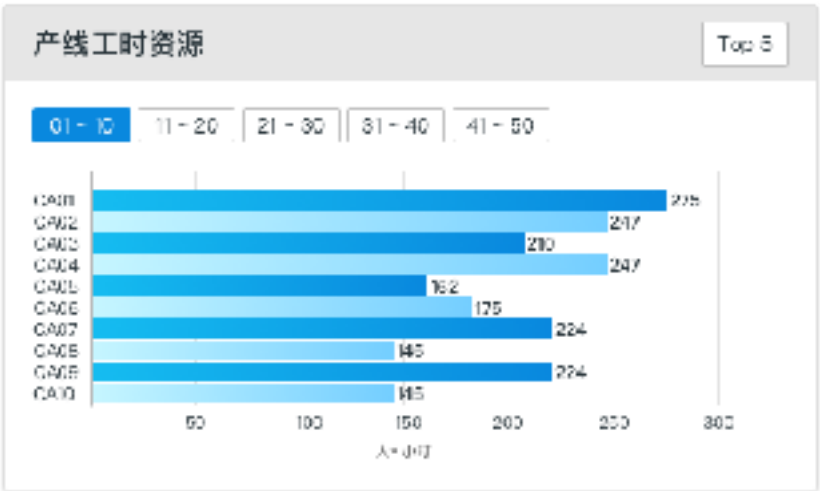
产品型号: 产品 AAA

数量: 100,000

单位: 个

确认

随工单号	产品编号	产品型号	数量	单位
S20230202020	P-212-224	产品 AAA	100,000	个
S20230202020	P-212-224	产品 AAA	100,000	个
S20230202020	P-212-224	产品 AAA	100,000	个
S20230202020	P-212-224	产品 AAA	100,000	个
S20230202020	P-212-224	产品 AAA	100,000	个
S20230202020	P-212-224	产品 AAA	100,000	个
S20230202020	P-212-224	产品 AAA	100,000	个
S20230202020	P-212-224	产品 AAA	100,000	个
S20230202020	P-212-224	产品 AAA	100,000	个
S20230202020	P-212-224	产品 AAA	100,000	个



随工单实时状态

Top 5

序号	随工单号	本单数量	未完成数量	完成耗时(h)	生产组	产品编号	生
1	S20230202020	99,000	9,000	124	CA01	CA1027-008-1452Z	C
2	S20230202020	100,000	1,000	22	CA02	CA1027-008-1452A	C
3	S20230202020	99,000	8,000	124	CA01	CA1027-008-1452Z	C
4	S20230202020	100,000	1,000	11	CA02	CA1027-008-1452A	C
5	S20230202020	100,000	0	124	CA02	CA1027-008-1452A	C

WAREHOUSE MANAGEMENT PAGE

NRE

资源管理

PLM 管理

库存管理

入库单

仓库

库存管理 / 仓库

仓库 Q-CS11111 明细

新增 删除

	仓库编号	仓库名称	仓管员	总数量	操作
☒	CH11111	Q-CS11111	TE	6,666	查看明细
☐	CH11112	A-CS11112	小明	1,122,334	查看明细
☐	CH11113	B-CS11113	小华	9,990	查看明细
☐	CH11114	CC-CS11114	小王	30,000	查看明细
☐	CH11115	C-CS11115	Mary	4,506	查看明细
☐	CH11116	C-CS11116	小明	12	查看明细
☐	CH11117	A-CS11117	小明	339,405	查看明细
☐	CH11118	CC-CS11118	Austin	49,506	查看明细
☐	CH11119	C-CS11119	小王	886	查看明细
☐	CH11120	C-CS11120	小王	999	查看明细

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Virtual Warehouse List

新增仓库

仓库编号MVP 开发 - 生产车间数字化平台功能示例

仓库名称

仓管员

排序编号

备注

取消 确定

Add Warehouse

查看条码

G123-123-123
专用 EL 变压器

A12345678A

关闭

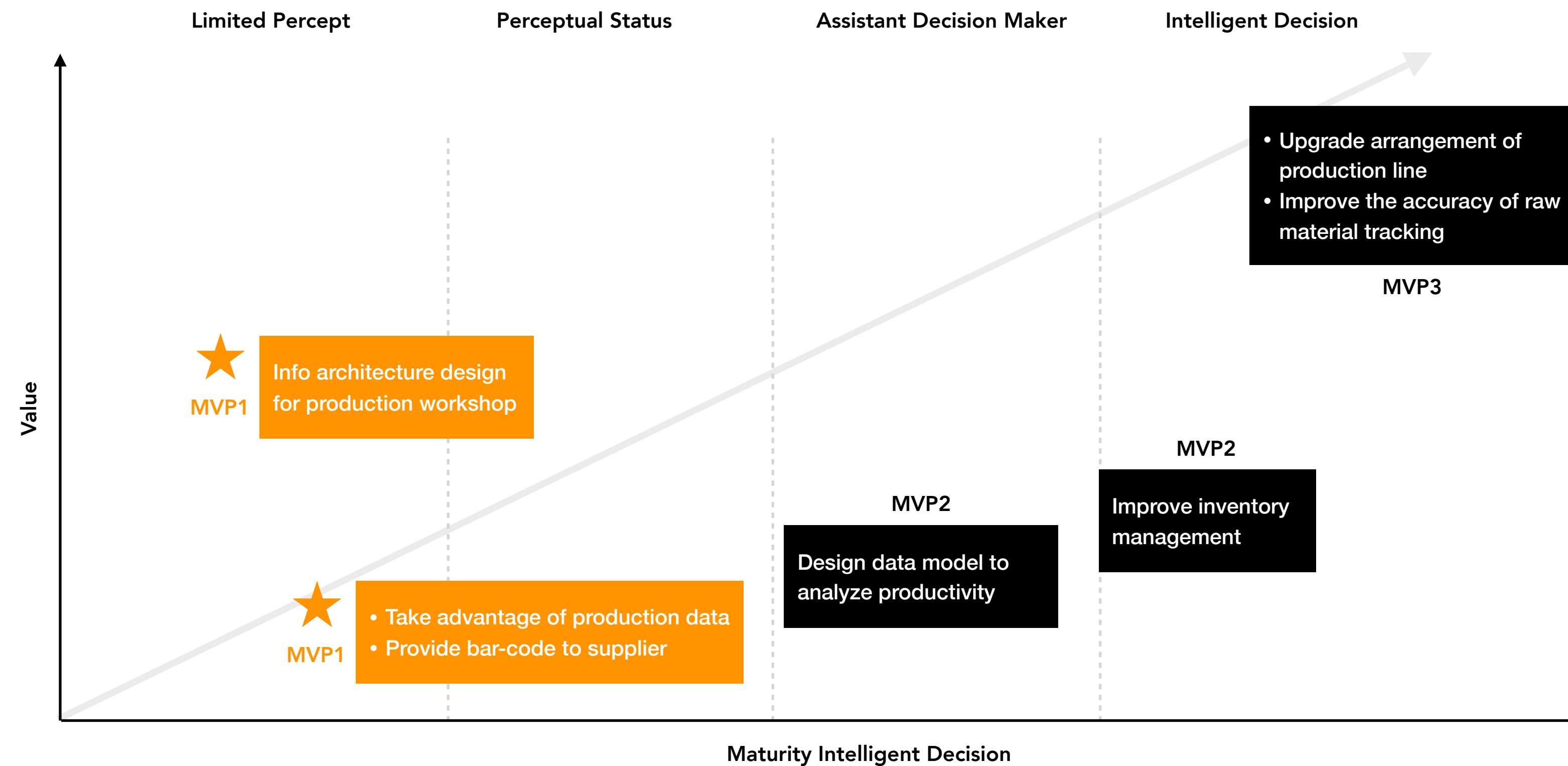
Warehouse Bar Code

Warehouse management page can help users manage finished products, semi-finished products and raw material classification, monitoring material status etc, to help managers more efficient manage resources in warehouse.

- Each warehouse will be assigned with an unique code, which can be classified more effectively.
- Each batch of product and sub-categories can be recorded in each virtual warehouse, which also can match the purchase order number
- According to material shortage alerts in platform, managers can make purchasing plan in advance.

THE VALUES : SMART FACTORY

MVP DEVELOPMENT ROADMAP



The Values of MVP 1

1. To help clients collect the data via equip IoT devices and 5G in production workshop.
2. Integrate the data, platform and productivity prediction.
3. Real-time labor and salary calculation.